AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 7 · STUDENT LESSON

Rows Columns and Data Types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Read the contract. Check the record. Limit the claim.

Distinguish field names, exact types, units and boundaries before documenting a source-backed repair.

Keep one habit: Duration contract → separate book audit → independent shelf schema → evidence limits.

MAIN LESSON · SECTION 1

Can a tidy table still contain a wrong fact?

> Specify fields, types, units and limits before judging records.

Your three goals

I can specify a table **schema**.

I can check **type**s and **unit**s as well as names.

I can distinguish structural validity from truth.

Recall contracts and debugging

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Readiness: name a field lookup

- What does row["used"] mean?

▼ Earlier wording, alternatives and picture transcripts

Your learning target

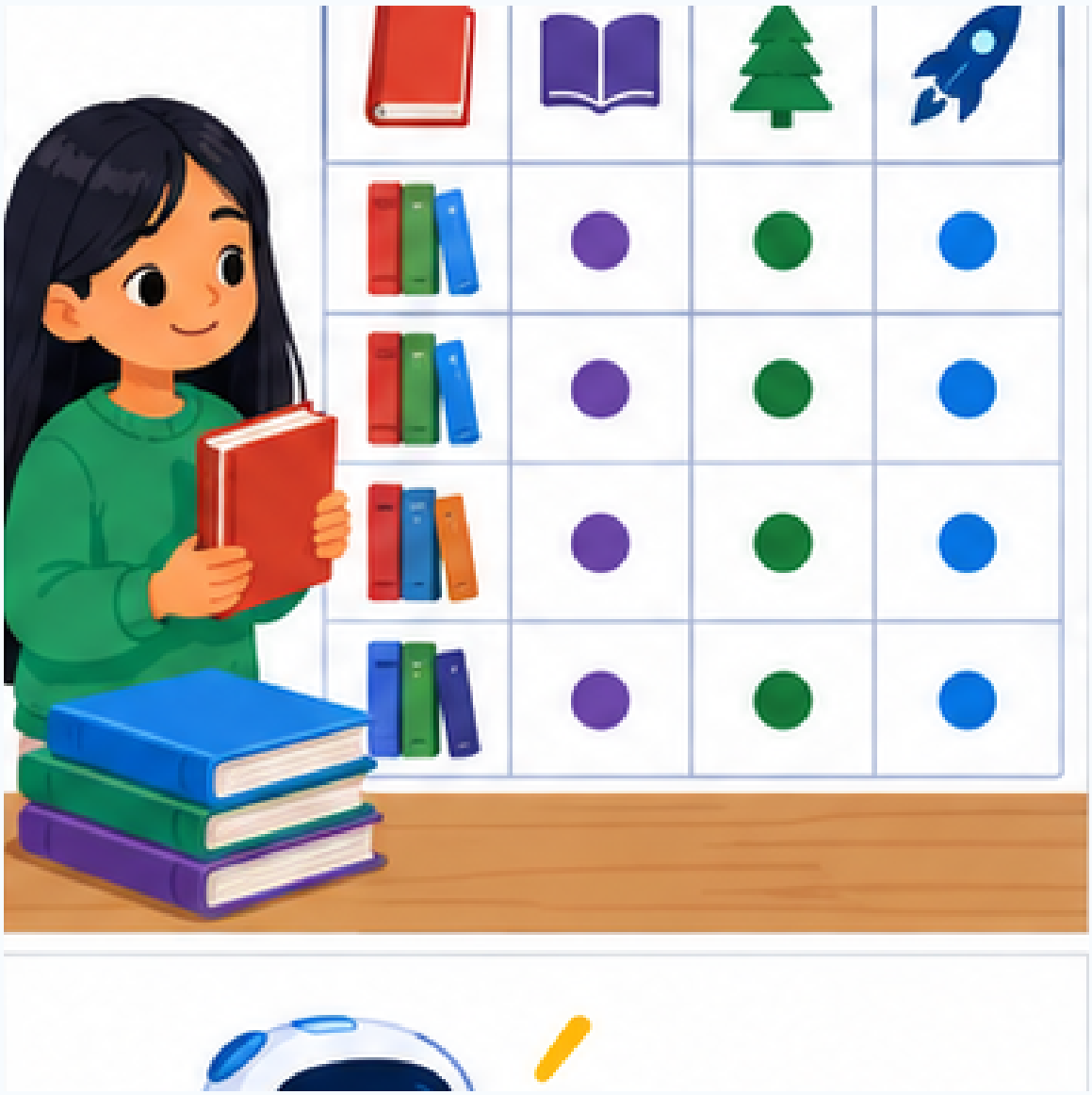
Remember before reading

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

Week 7: Rows, **Column**s, and **Data Type**s

Structured tables preserve meaning through named fields and consistent **type**s.



THINKER MISSION

Repair a mixed-up class-pet dataset.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

By the end, I can read a table **schema** and distinguish a numeric value from its text representation.

MAIN LESSON · SECTION 2

Read the activity-duration contract

> Use the same meaning and unit in every row.

One session per row

A fictional classroom logs the duration of each activity. Before comparing durations, define a **schema**: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session.

These decisions make data interpretable.

What the contract does and does not require

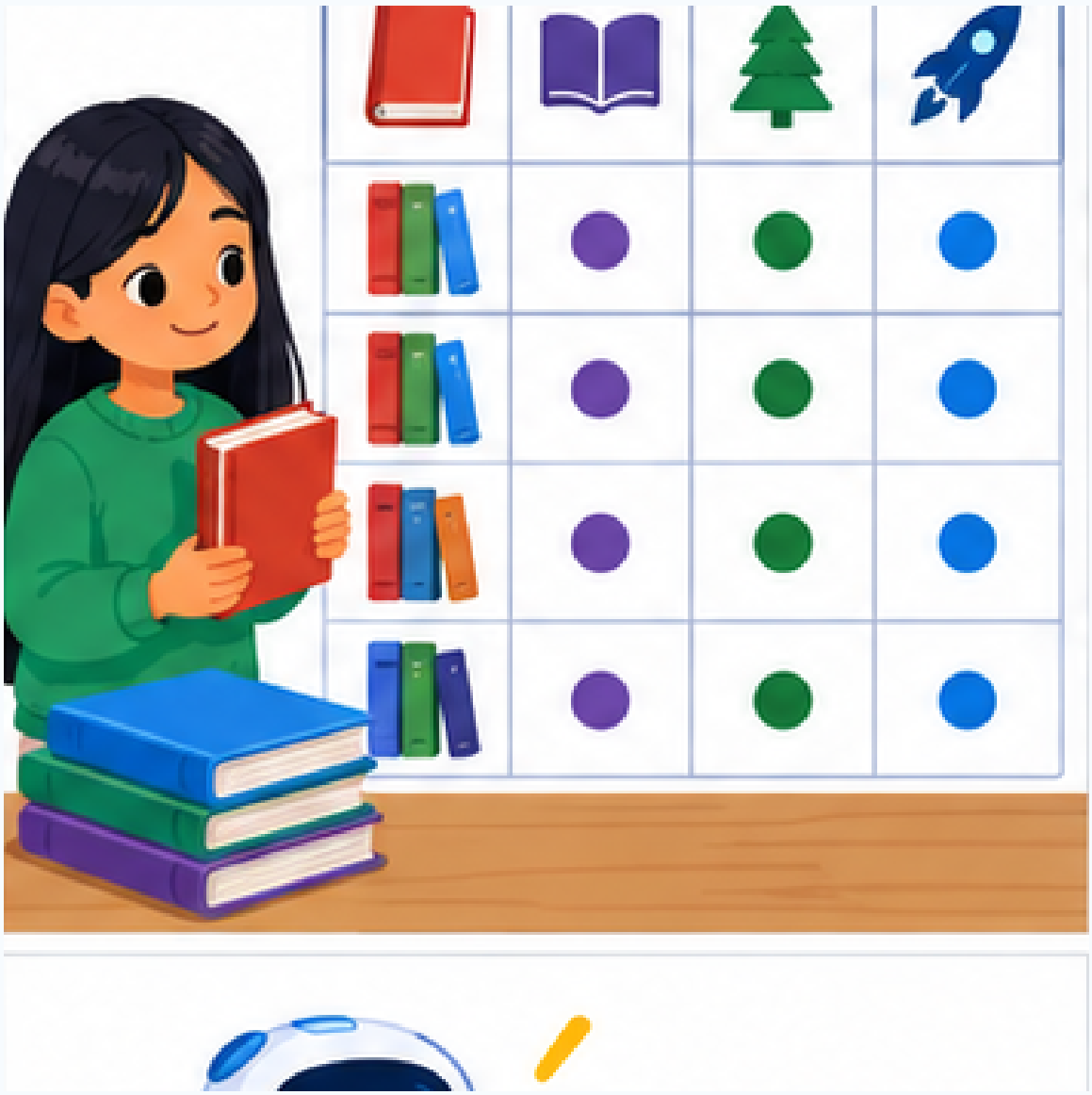
A row represents one session; a column represents the same named field across sessions. A schema is the table’s written contract. Here minutes means whole minutes, including 0 and 60. Nonempty means at least one character in this simple example; a spaces-only name would pass the code, so a stricter real application needs an explicit whitespace rule.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use rows, **column**s, and **data type**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Check presence, type and range separately

> The supplied Python example assumes both fields exist.

Read the complete validator

```
rows = [{"activity": "Reading", "minutes": 12},
        {"activity": "Sorting", "minutes": 0}]
for row in rows:
    valid = (type(row["activity"]) is str
             and len(row["activity"]) > 0
             and type(row["minutes"]) is int
             and 0 <= row["minutes"] <= 60)
    print(valid)
```

Explain each check

A **schema** is a checklist for what a row must contain. First ask whether the activity name is text and is not empty. Next ask whether minutes is an integer.

Then ask whether that integer lies between 0 and 60, including both ends. Keep these checks separate when explaining a rejection.

The number 12 and the text "12" look similar on a page, but only the number meets this lesson's minutes rule.

The worked script prints True twice. **type**(value) is int checks the exact **type** here and avoids accepting a Boolean as a count. The chained comparison checks both boundaries. The example assumes both keys exist; this week's paper audit identifies missing fields without treating this script as a complete missing-field validator. A zero duration is allowed by this written **schema**.

The schema at a glance

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
activity	Nonempty text	Names the session activity
minutes	Integer 0 through 60	Duration in minutes

Read each row from left to right and connect the evidence to the stated rule.

Connect a diagram to a deciding check

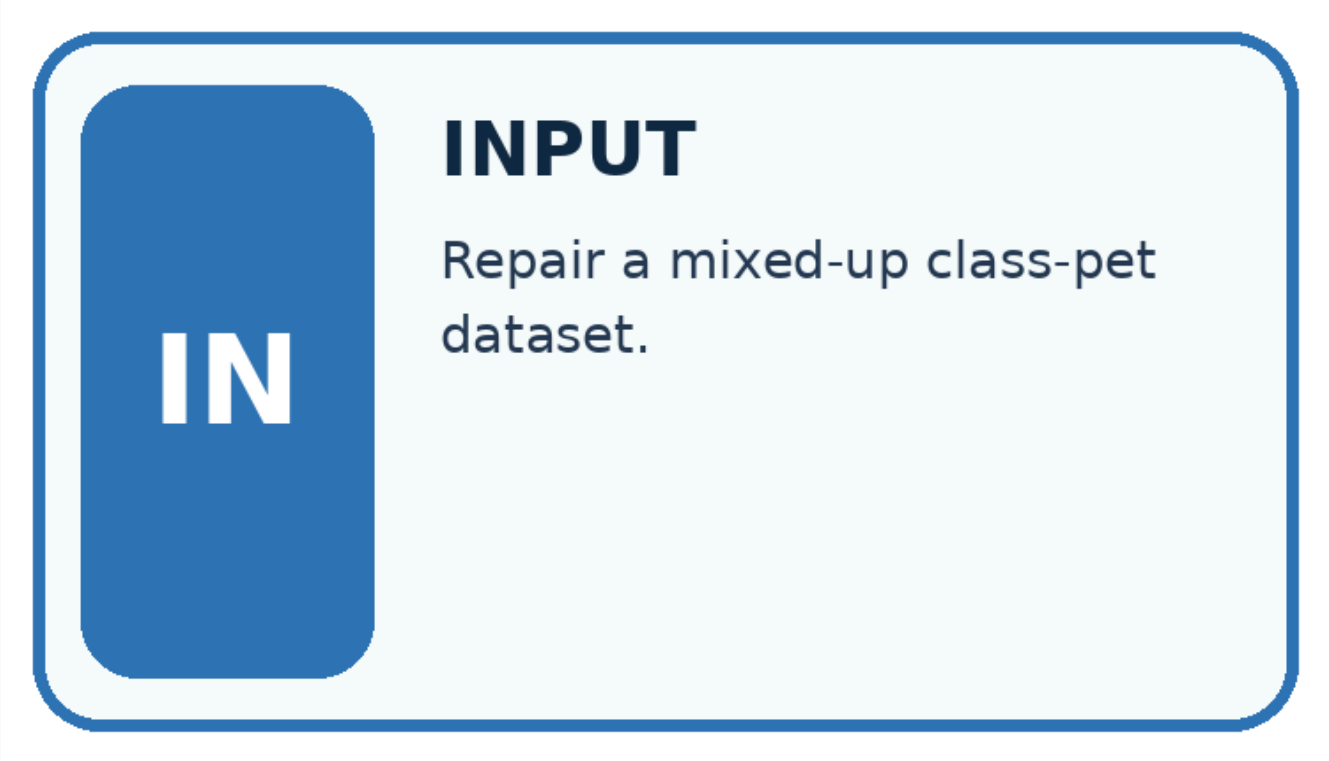
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

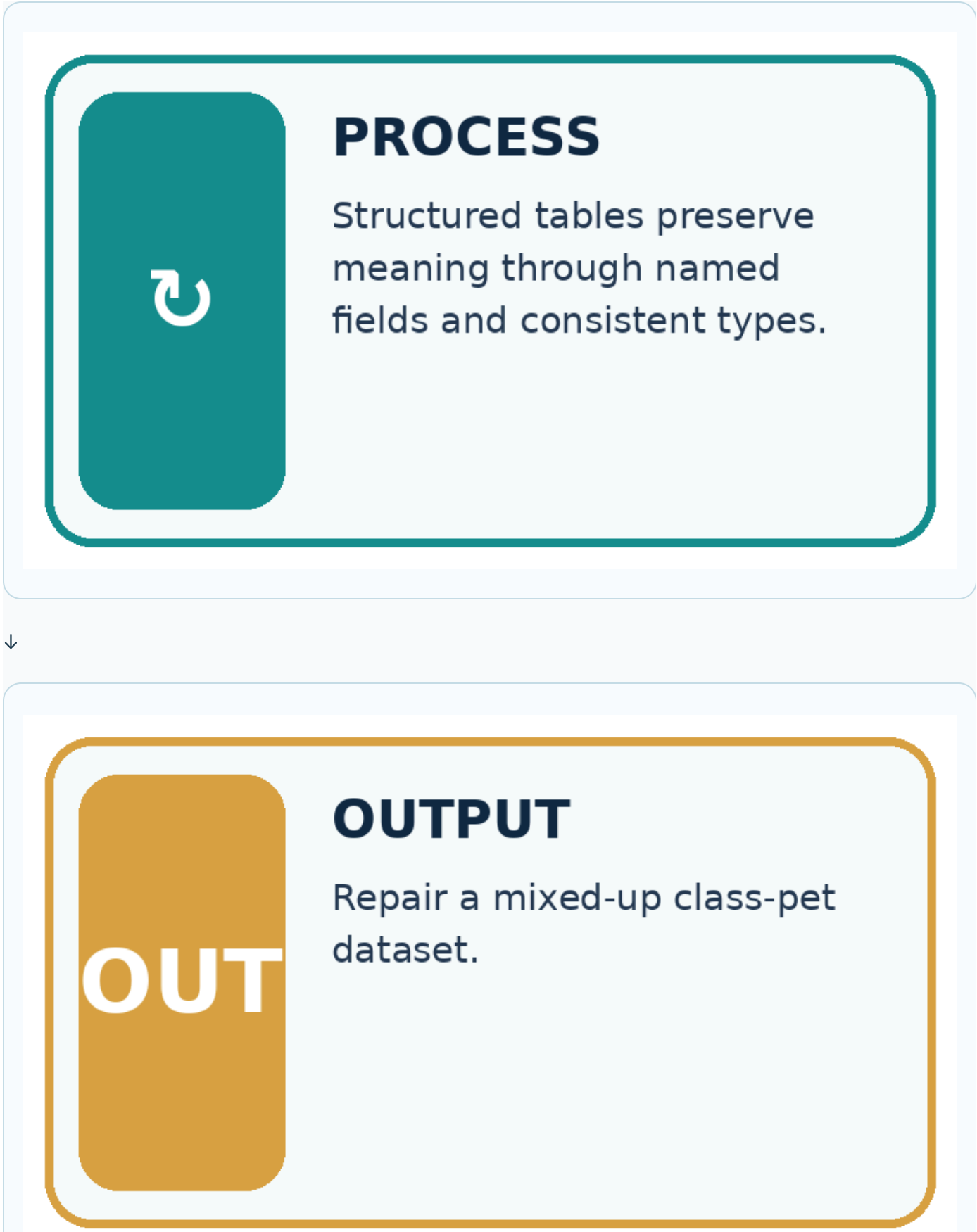
VISUAL **MECHANISM**

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


↻
PROCESS
Structured tables preserve meaning through named fields and consistent types.
↓
OUT
OUTPUT
Repair a mixed-up class-pet dataset.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Check a row in separate steps

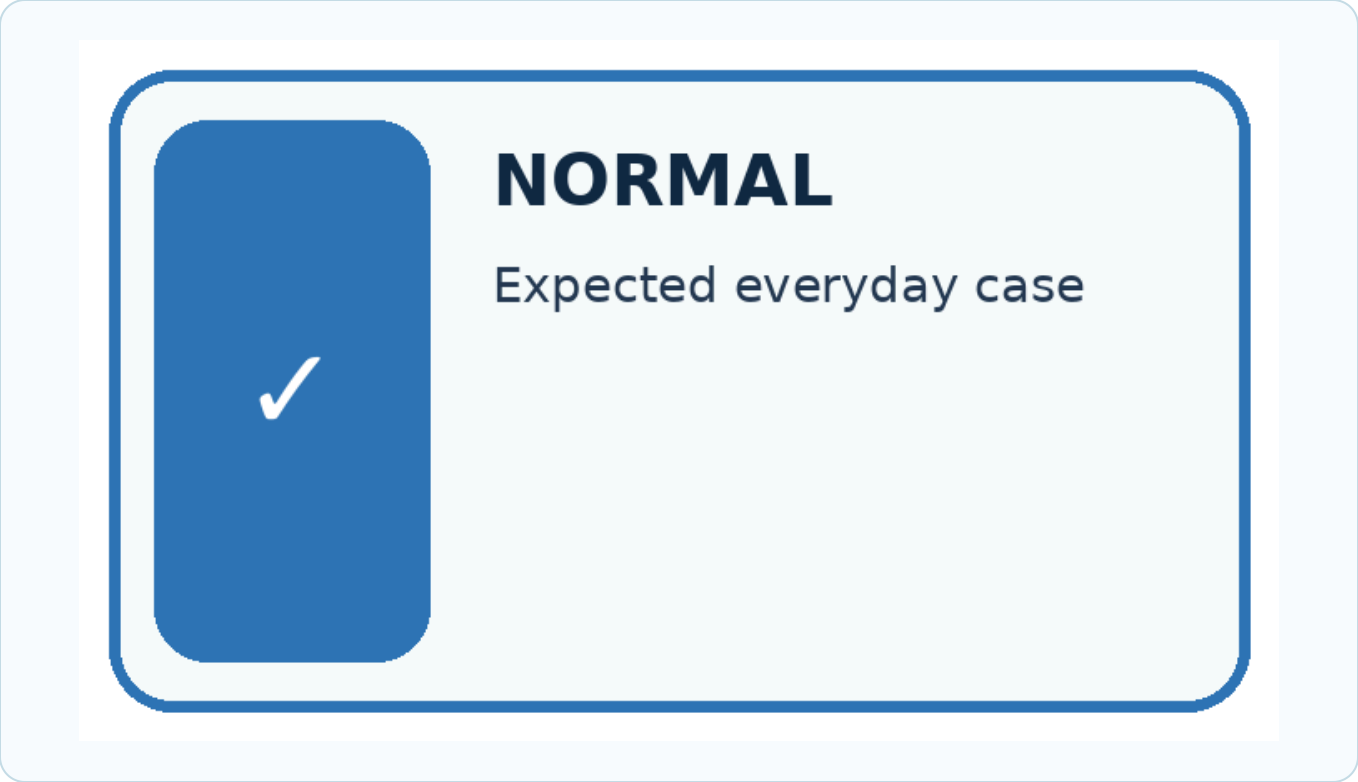
Connect the picture and the explanation

MAIN LESSON · SECTION 4

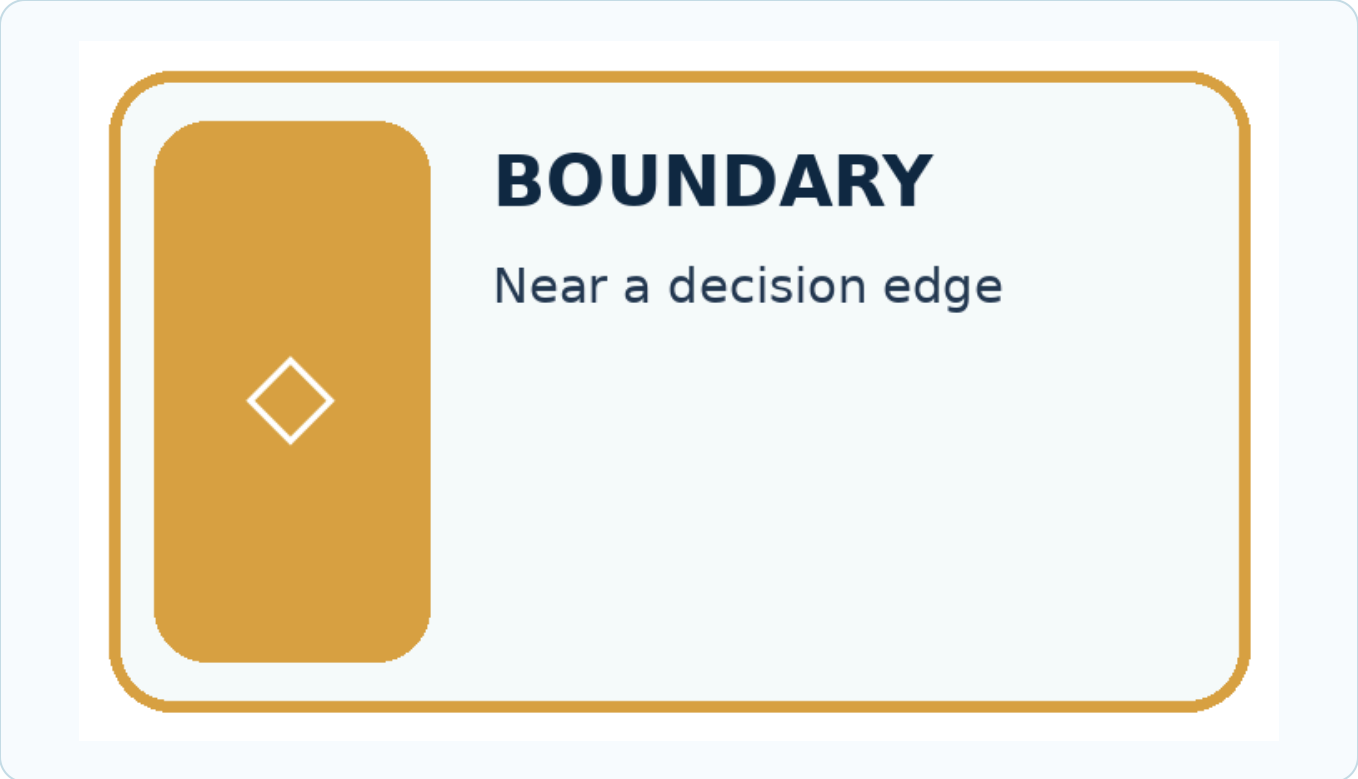
Choose a revealing record

> An ordinary pass, a boundary and a type error test different rules.

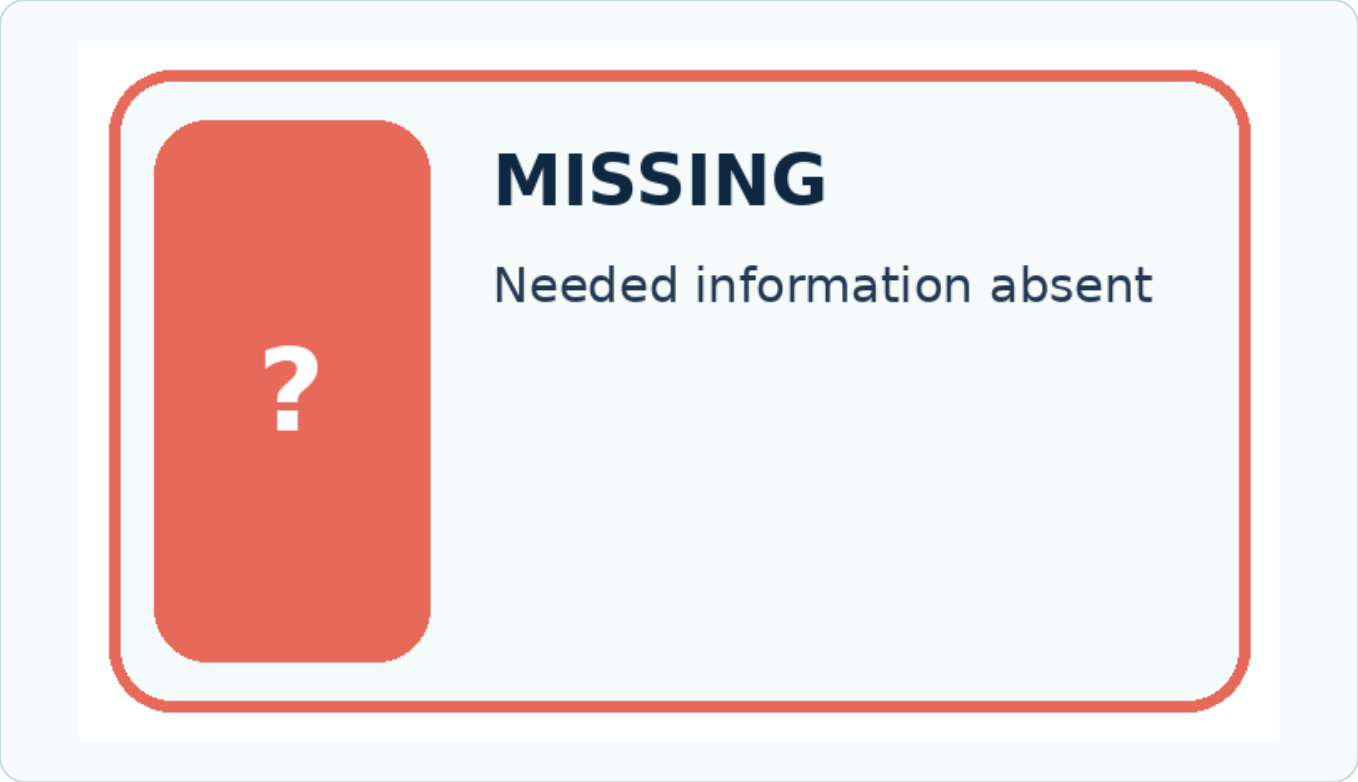
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Change B's pages from "90" to integer 90 using an approved repair.

▼ Check after predicting

Prediction: B now satisfies the declared **schema**.

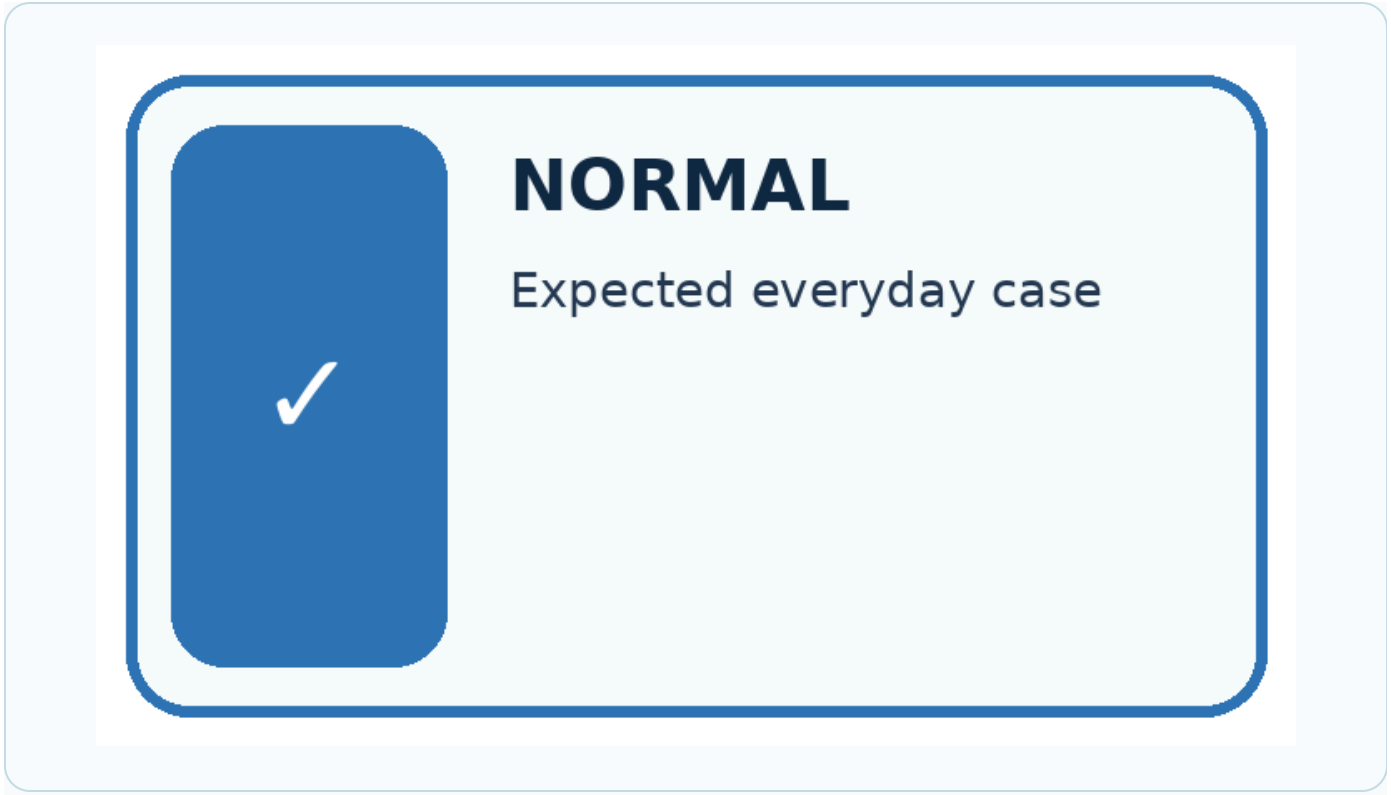
Reason: Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

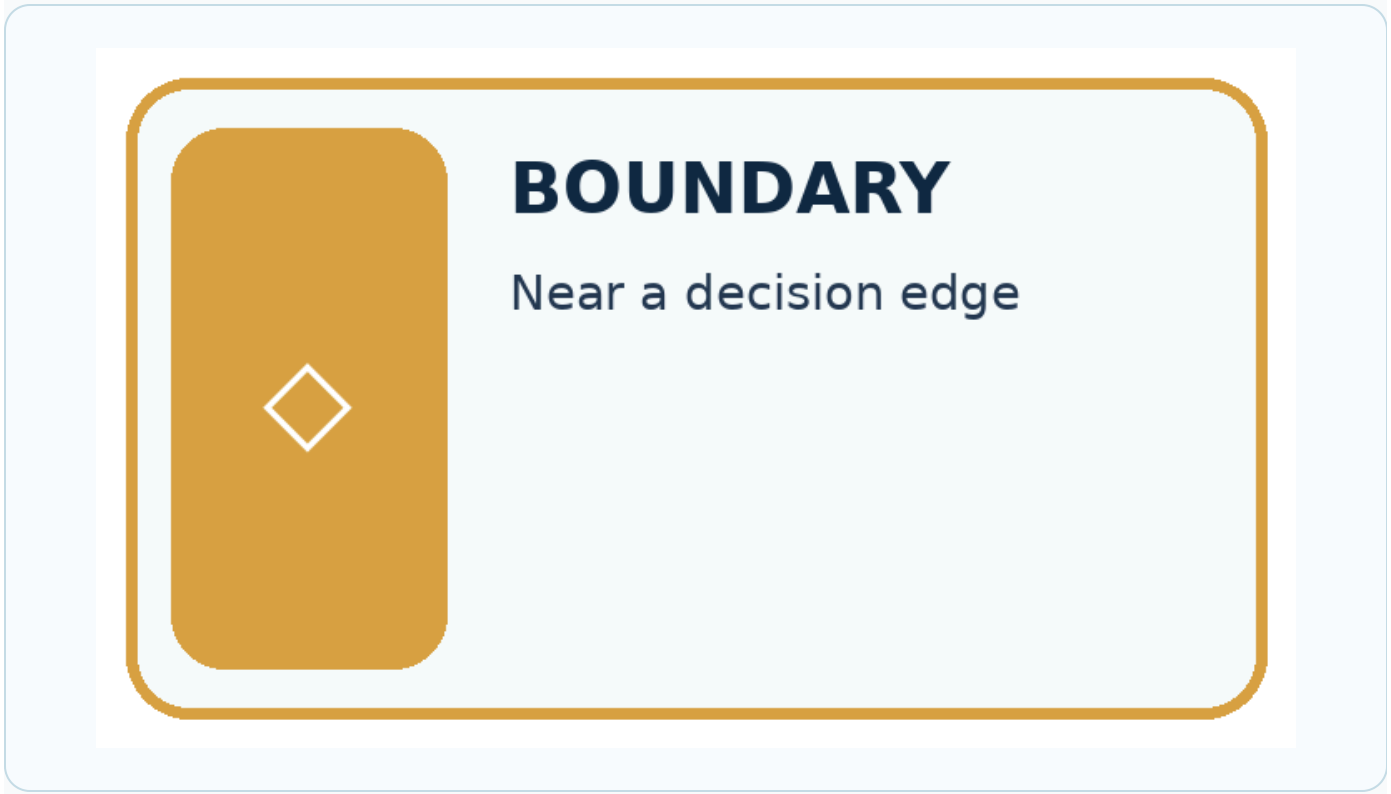
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

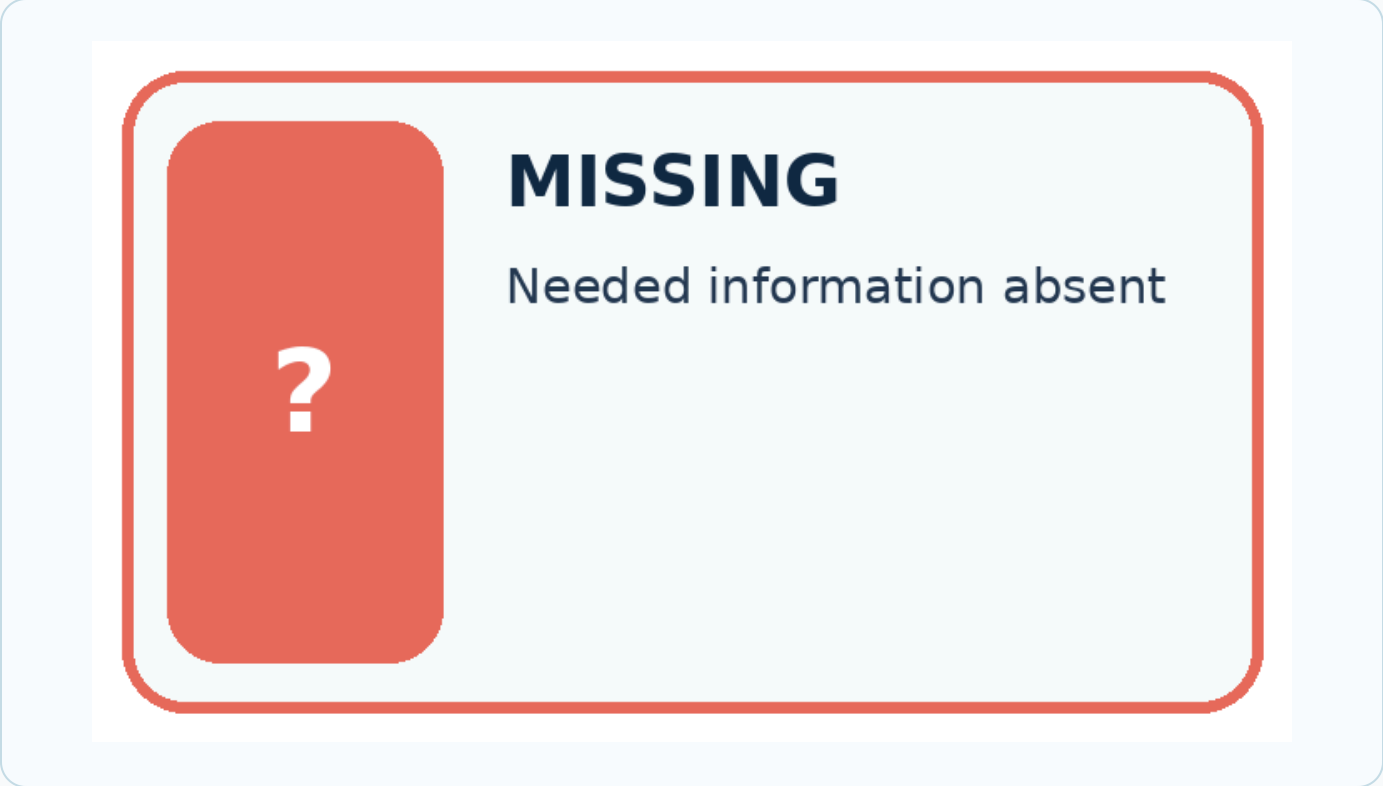
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

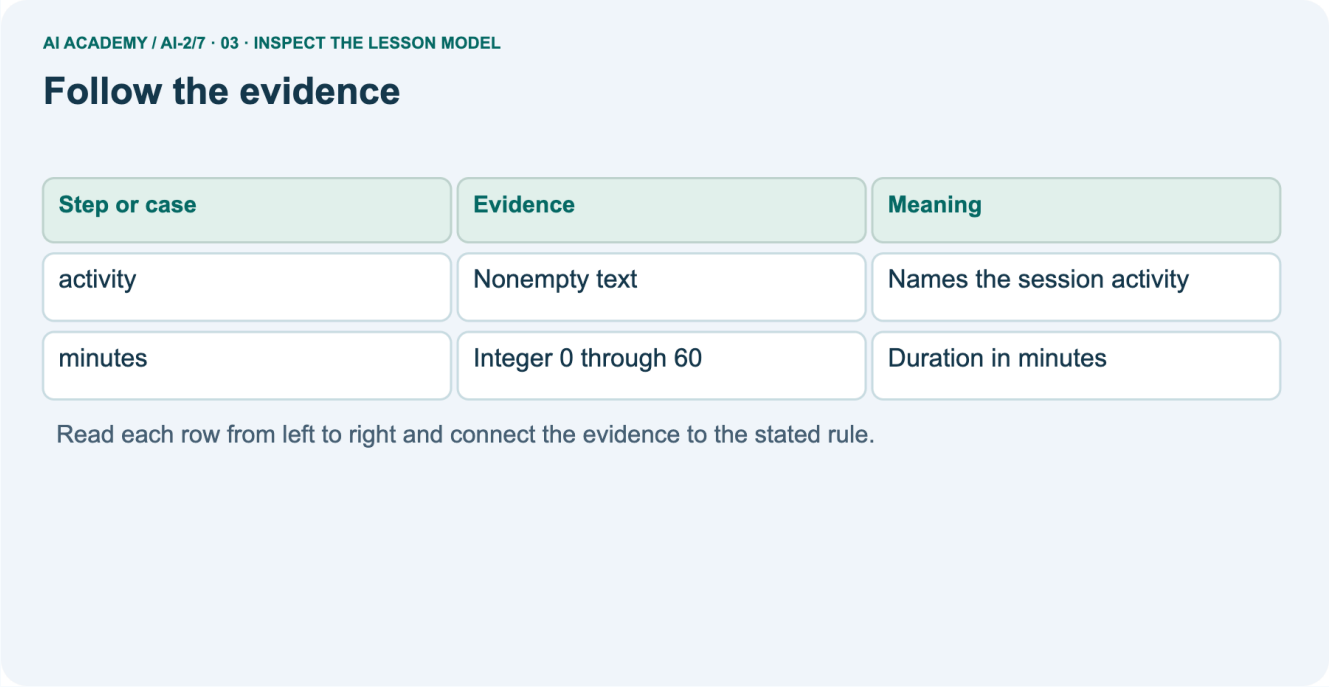
Compare duration checks with a separate book schema

> Do not carry duration boundaries into page counts.

Predict the duration results

Would minutes of 61, "12" and 12 each pass? Explain the type distinction.

The duration validator picture



Follow the labelled picture with the matching explanation. Open the vector diagram

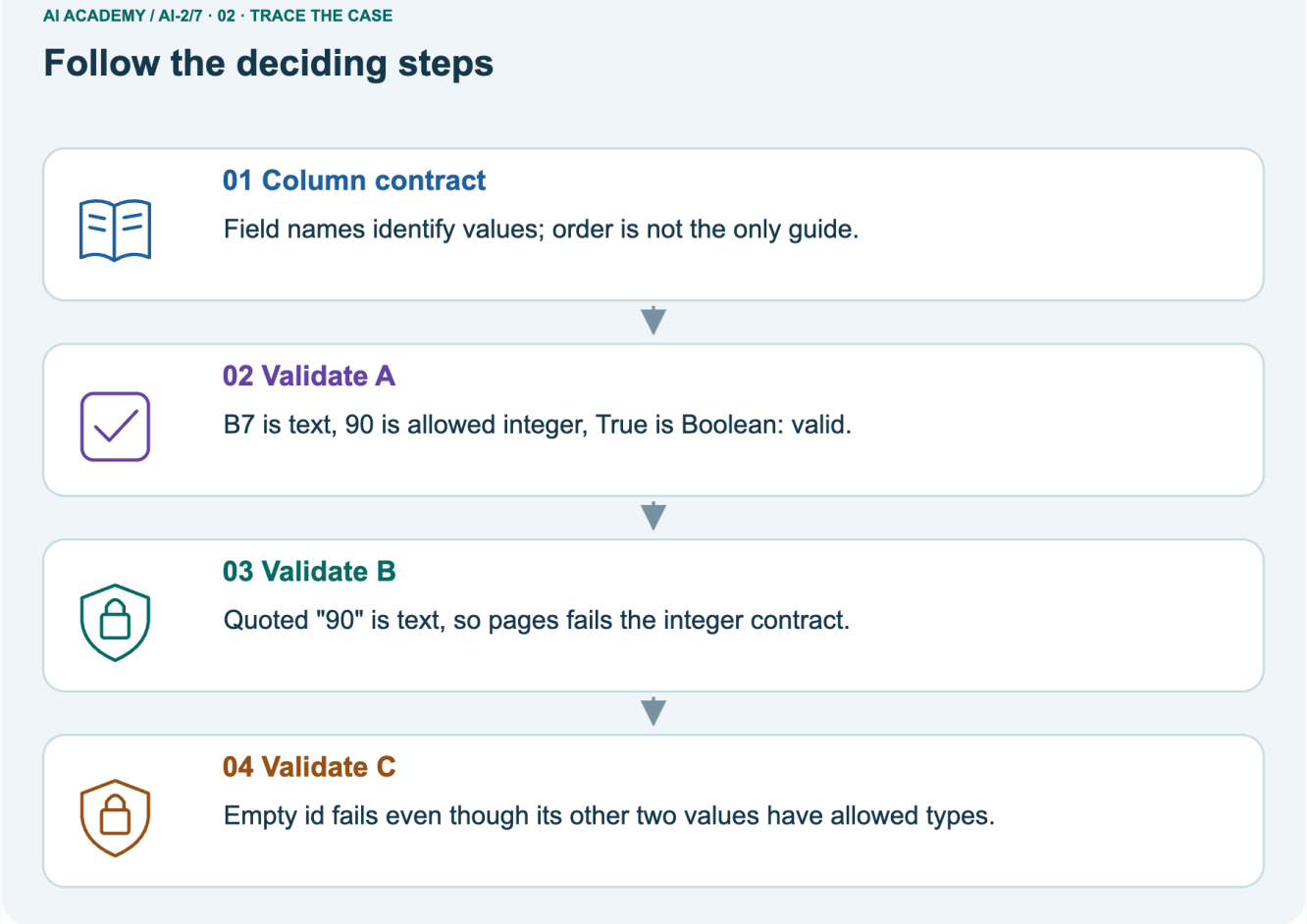
Introduce the separate catalogue rules

A fictional approved book catalogue has three fields per row.

Schema: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False}; C={id:"",pages:80,illustrated:True}.

Trace A, B and C against those rules



Follow the labelled picture with the matching explanation. Open the vector diagram

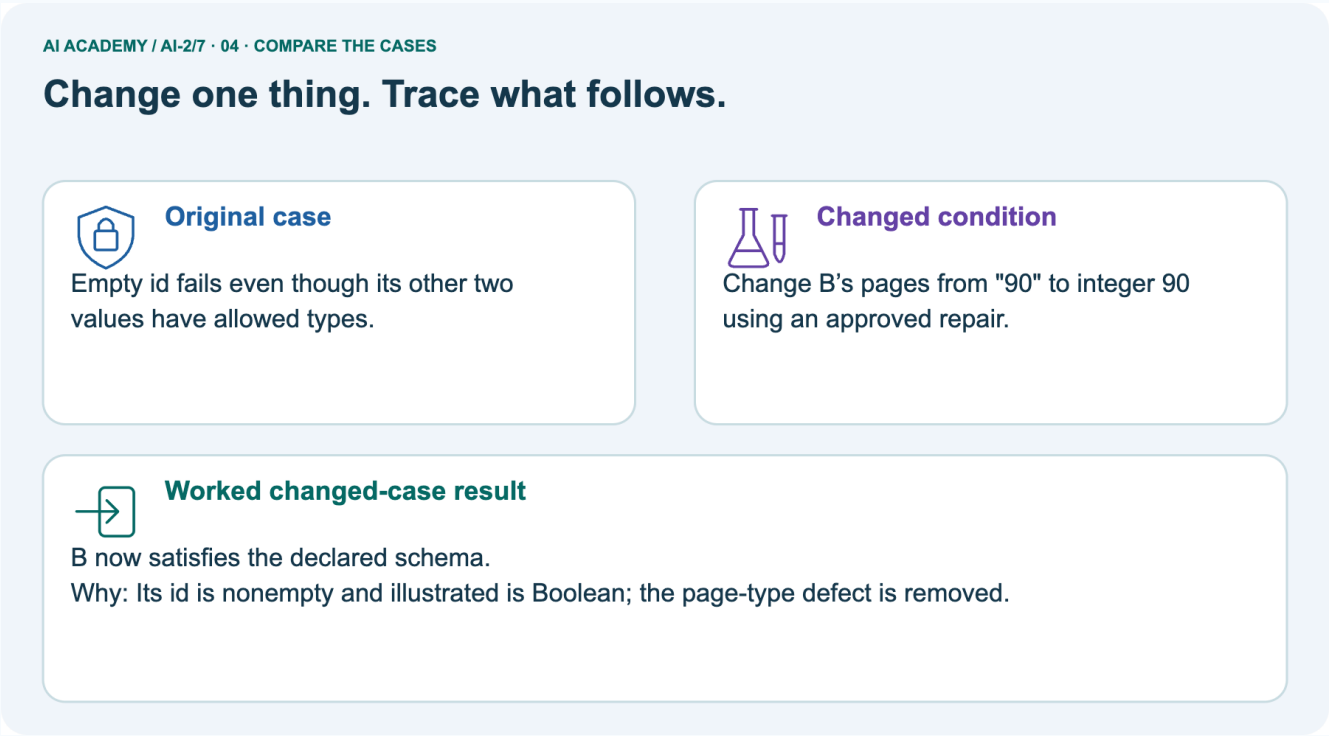
A **schema** makes row meanings and expected **type**s explicit. The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source.

Predict one authorised repair

- Change one thing: Change B's pages from "90" to integer 90 using an approved repair. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain failures and the truth limit

- Read the labelled picture: Which two rows originally fail, and do they fail for the same reason?

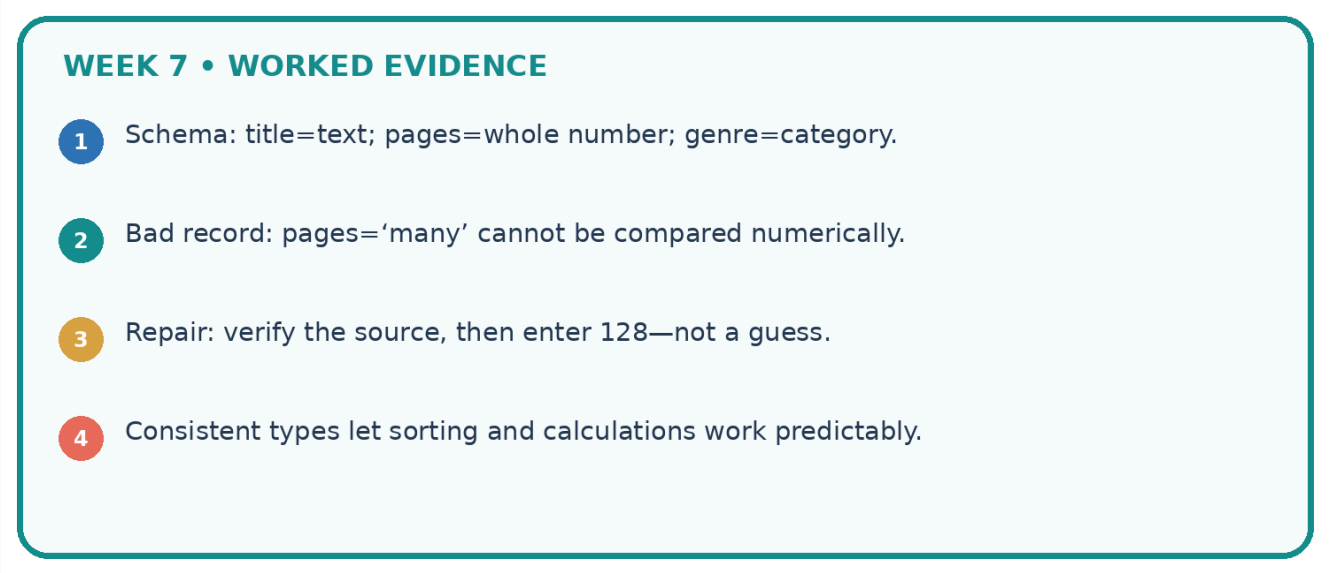
- **Schema** validity does not confirm that the book really has 90 pages or that its metadata is accurate.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Schema**: title=text; pages=whole number; genre=category.

2. Bad record: pages='many' cannot be compared numerically.

3. Repair: verify the source, then enter 128—not a guess.

4. Consistent **type**s let sorting and calculations work predictably.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

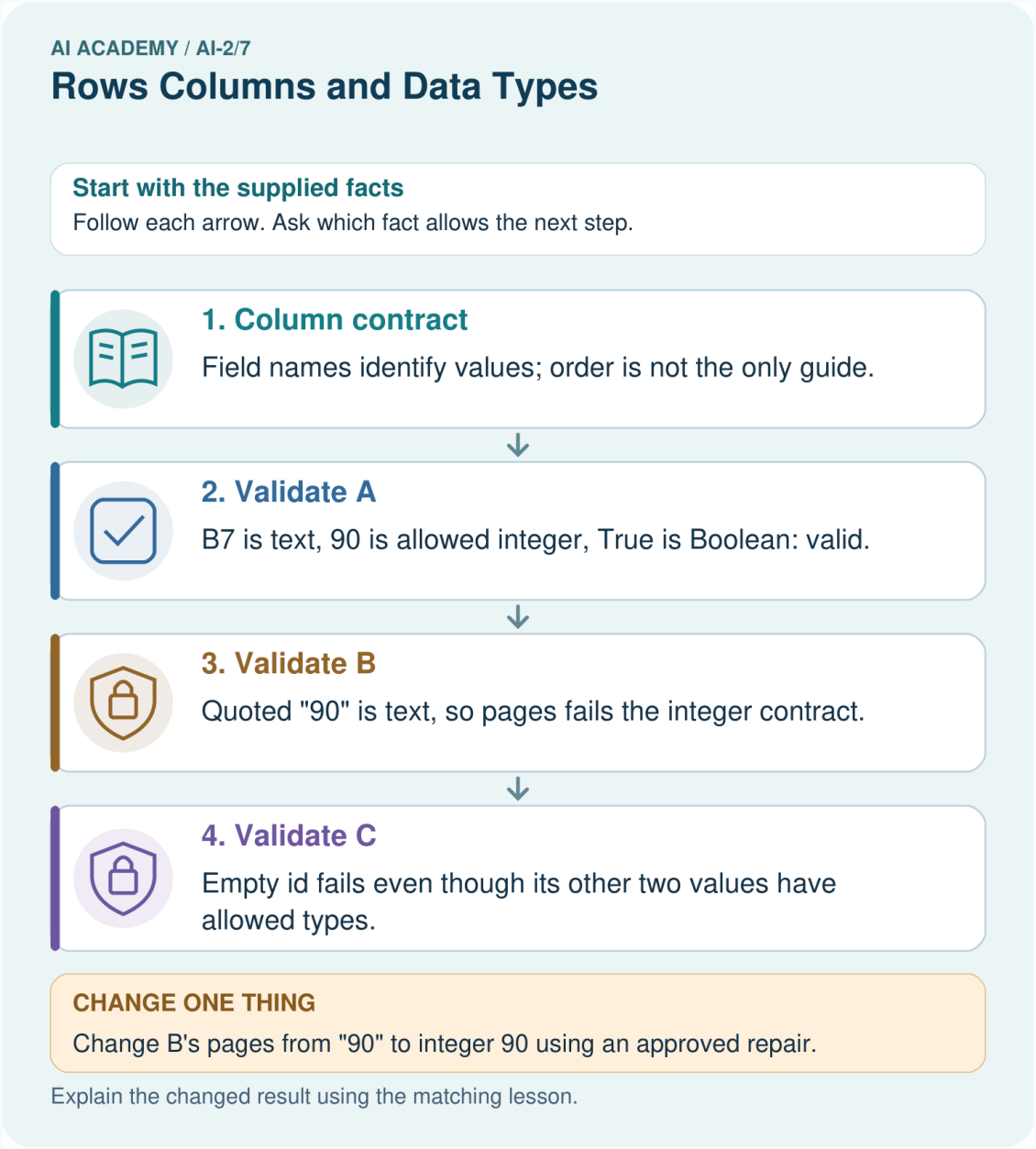
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Rows Columns and Data Types

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Column contract	Field names identify values; order is not the only guide.
2. Validate A	B7 is text, 90 is allowed integer, True is Boolean: valid.
3. Validate B	Quoted "90" is text, so pages fails the integer contract.
4. Validate C	Empty id fails even though its other two values have allowed types.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Separate valid structure from a true observation

> A permitted value still needs trustworthy collection evidence.

What a successful check establishes

A row can satisfy every **schema** rule and still be a wrong **observation**. **Schema** validation checks our stated structure and **constraint**s, not whether someone measured honestly or selected a representative sample.

Challenge the measured-duration claim

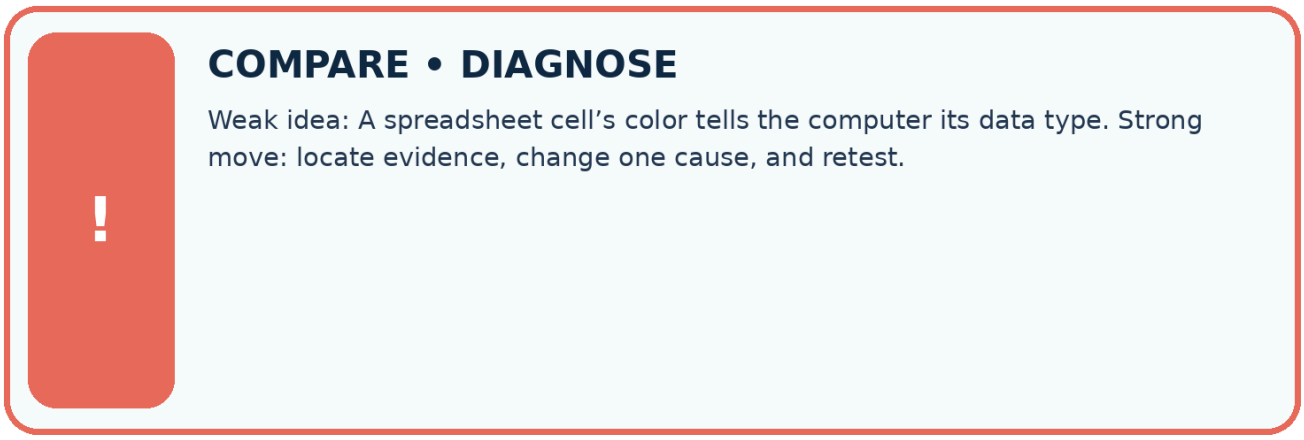
> **A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 8

Practise boundaries, units and justified conclusions

> A new label does not convert a measurement.

Repair a seconds-to-minutes mismatch

If one row records 120 seconds in the minutes field, a simple numeric total becomes misleading. Correct the source or explicitly convert and document it; renaming a **column** alone does not convert values.

Classify the workbook records

Use workbook W2 for the connected example “Rows **column**s and **schema**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge a successful-run guarantee

- Use the worked case for Rows **column**s and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the separate catalogue failures

Which two rows originally fail, and do they fail for the same reason?

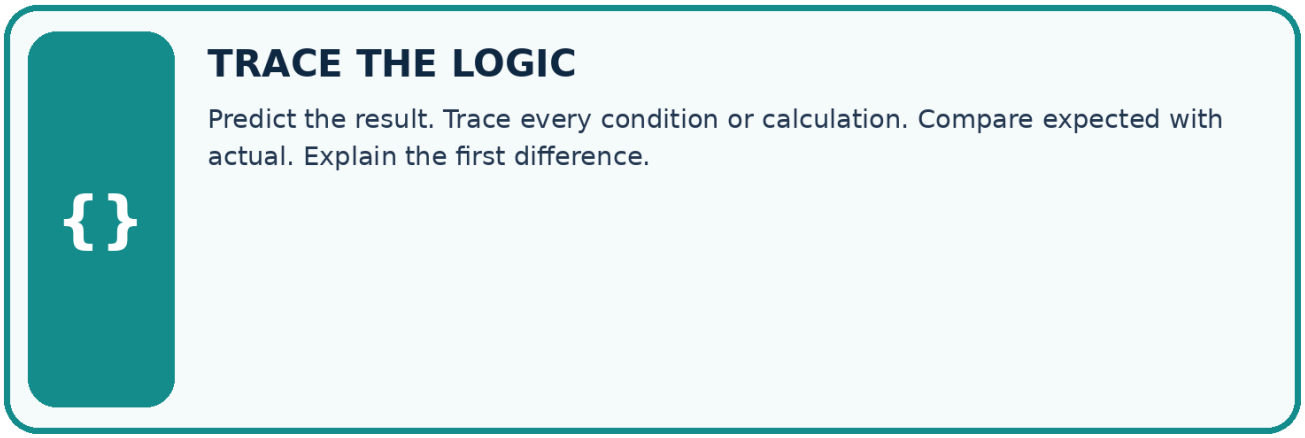
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Classify records

Check your reasoning

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Audit a supplied ten-row catalogue

> Keep schema failures, repeated identifiers and uncertain facts separate.

The exact audit contract

Use this fictional practice catalogue only. Required fields: id is nonempty text; pages is an exact integer from 1 through 300; illustrated is Boolean True or False. Every row describes one book. Keep quotes visible: "90" and "false" are text, while 90 and False are different types. A duplicate id is an audit flag for checking the source, not a type error or permission to delete a row. Do not invent corrections.

The ten records

Row	id	pages	illustrated
1	"B1"	90	True
2	"B2"	"90"	False
3	""	80	True
4	"B4"	0	False
5	"B5"	300	False
6	"B6"	301	True
7	"B7"	120	"false"
8	"B1"	90	True
9	"B9"	missing	False

Row	id	pages	illustrated
10	"B10"	1	False

Predict, audit and document

1. Before checking, predict each row's result. Audit by hand: mark missing fields first, then type, then allowed range. Record the first reason and any additional problem. Flag the repeated B1 separately. For every proposed repair, state the source evidence required; leave unresolved values unresolved. The supplied Python duration script is not a complete catalogue validator and can raise KeyError on missing fields. Label this investigation a paper audit unless you implement and execute a matching validator.

Keep the evidence and next question

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____

> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → TEST → RECORD → EXPLAIN

Audit a 10-row book table. Circle mixed types, missing values, duplicated rows, and fields with unclear names; document each repair.

Audit a 10-row book table. Circle mixed **type**s, missing values, duplicated rows, and fields with unclear names; document each repair.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: design the shelf-audit schema

> Give your own records before consulting feedback.

The new field contract

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

Your first attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Design a schema independently

Week 7 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A table row is book_id B7, pages 84, available true. Name the **type** of each field and explain why pages stored as eighty-four complicates calculation.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Store one book per row and keep reader preferences in a separate record.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-**input** case.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Publish the recommender’s schema before ranking

> Keep catalogue rows separate from reader preferences.

The project checkpoint

THIS WEEK’S CHECKPOINT Store one book per row and keep reader preferences in a separate record.

Publish the recommender’s row **schema** and validation results before ranking books, preserving repair provenance.

Keep inputs, decisions and limits inspectable

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

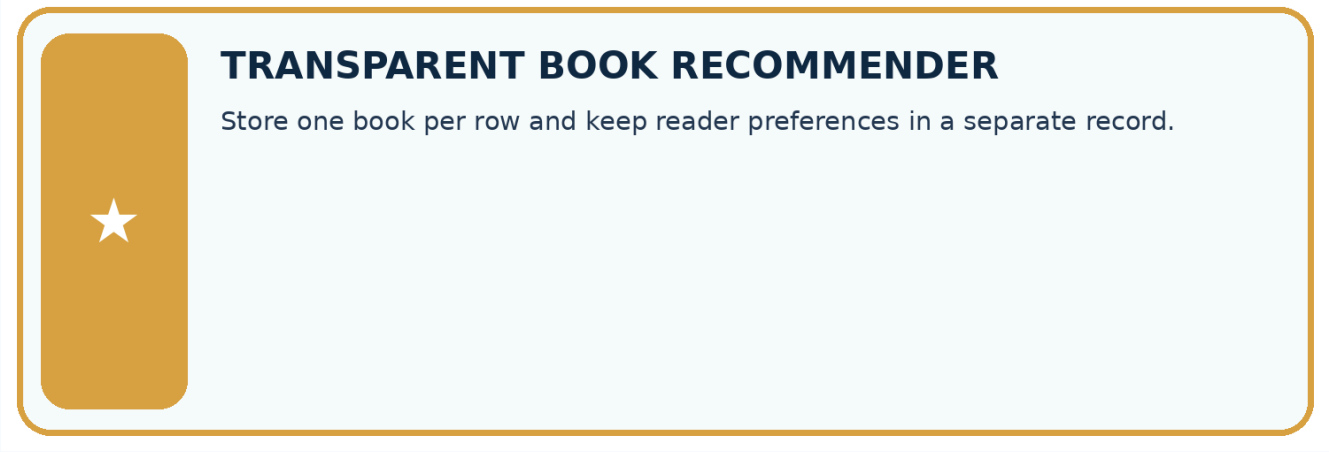
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain what a schema can establish

> Recall the contract and one remaining evidence limit.

Explain without notes

- Close your notes. Explain this goal in your own words: Specify a table **schema**

My deciding step, evidence and one remaining question

Choose a changed case for return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
row	One record or example in a table.	
column	A table field containing the same kind of information across rows.	
data type	A category of values, such as number, text or Boolean, with defined operations.	
schema	A specification of required fields, types and structural rules for data.	

Terms for the connected explanation

Word	Meaning
Type	A category such as integer or text
Unit	The quantity scale attached to a number

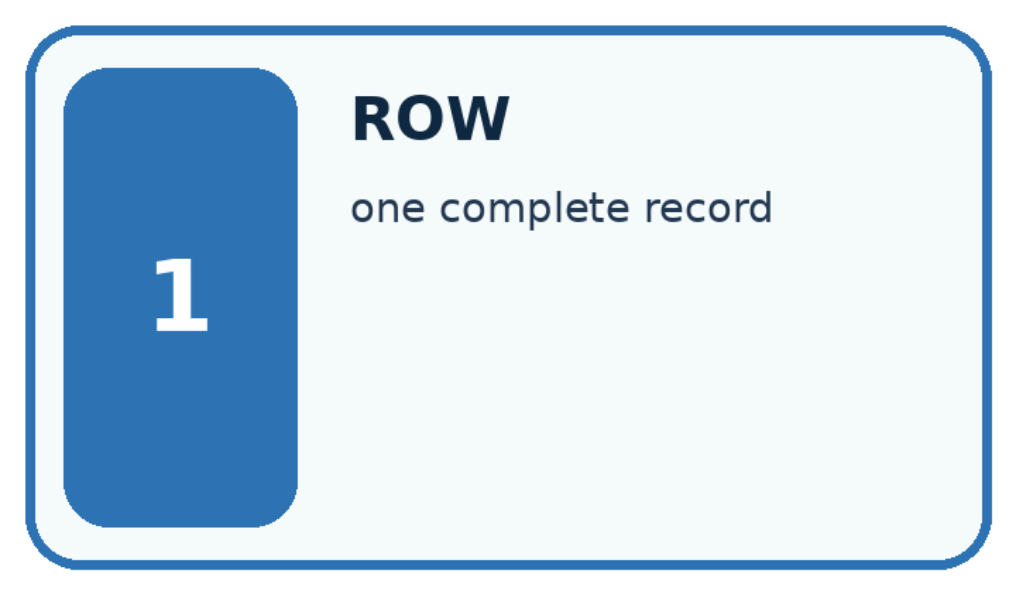
Word	Meaning
Constraint	A rule a valid record must satisfy

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<summary>Earlier wording, alternatives and picture transcripts</summary>

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



</details>

2

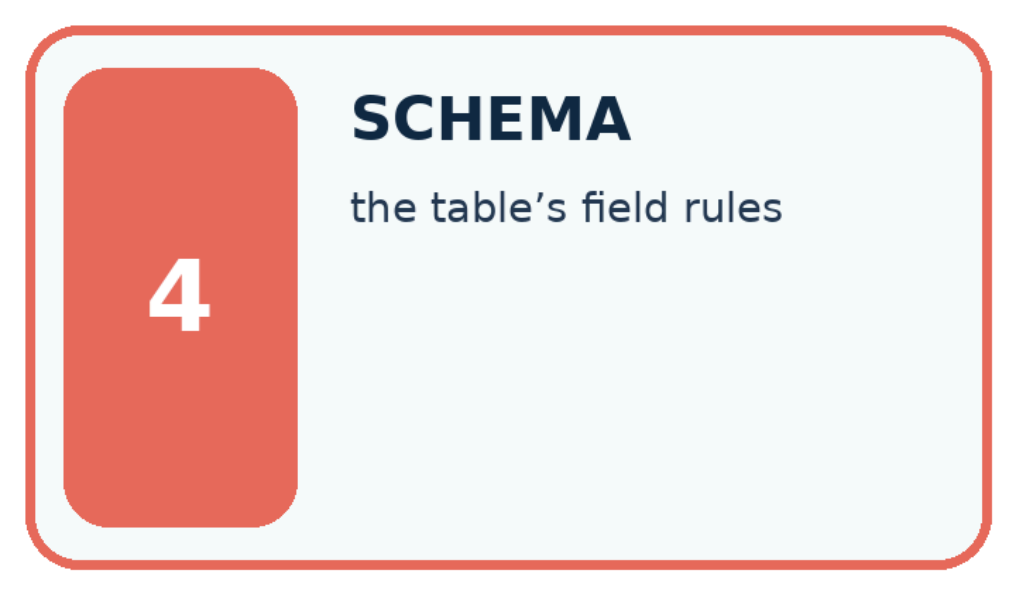
COLUMN

one named field across records

3

DATA TYPE

kind of value allowed



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

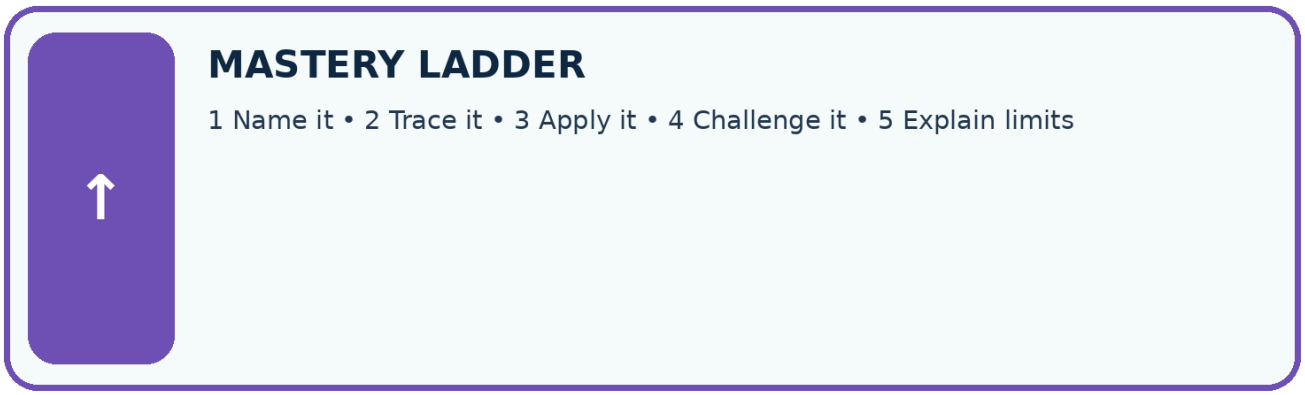
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Design a schema independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

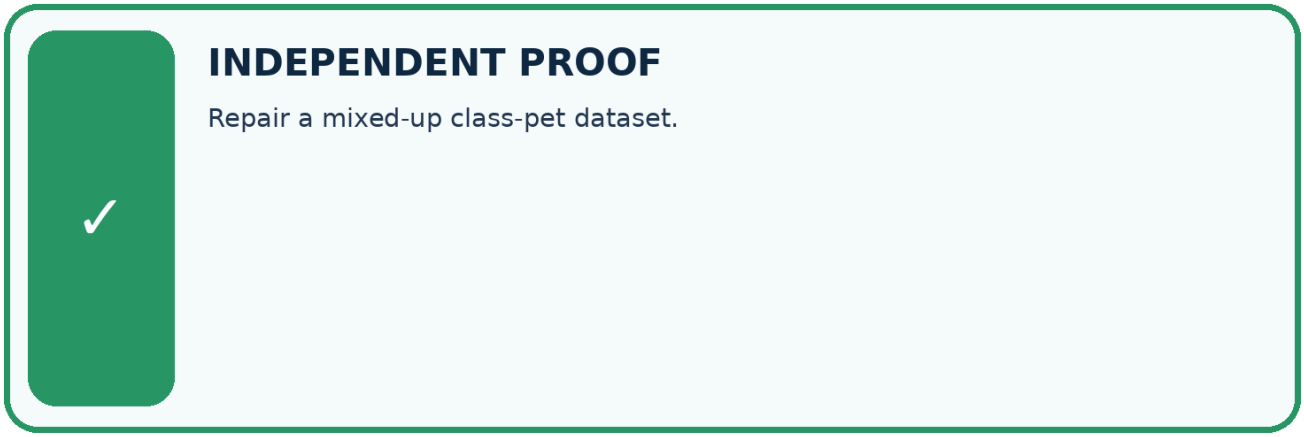
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Design a schema independently is the selected independent case.



EXPLANATION CHECK State field meanings, exact types, units and constraints; distinguish structural validity from factual accuracy.

Write or draw your reasoning and evidence.

▼ **Extra practice: A field has a meaning and a type**

REFERENCE · SECTION 13

Extra practice: A field has a meaning and a type

> Try the question before opening its worked explanation.

Read the supplied case and question

Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

Assign intended **type**s and a repair.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Text, numeric and Boolean. Confirm ninety means 90, then convert and record the change; use a consistent Boolean representation.

Evidence or stage	Value or result
Title	Text → River
Pages	Number → 90 after checking source
Borrowed	Boolean → true or false

▼ Earlier wording, alternatives and picture transcripts

Week 7 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A field has a meaning and a type

Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

Assign intended **type**s and a repair.

Text, numeric and Boolean. Confirm ninety means 90, then convert and record the change; use a consistent Boolean representation.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Specify a table schema	Explain the deciding step, show a trace or test, and state one limit.
Check types and units as well as names	Explain the deciding step, show a trace or test, and state one limit.
Distinguish structural validity from truth	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional approved book catalogue has three fields per row.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional approved book catalogue has three fields per row.
- Which labelled step matches this detail? Empty id fails even though its other two values have allowed **type**s.
- 1. **Column** contract
- 2. Validate C
- 3. Validate B
- 4. Validate A

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Validate C Empty id fails even though its other two values have allowed **type**s. A **schema** makes row meanings and expected **type**s explicit.

The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source.

Changed-case question

- Changed-case practice: Change B's pages from "90" to integer 90 using an approved repair. Which conclusion is supported by the supplied facts?

- 1. B now satisfies the declared **schema**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: B now satisfies the declared **schema**. Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

Repair and save your work

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?.

Return to this week and explain what you changed.

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If print(total) moves before total=total+2, what gets printed for the original **input**?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

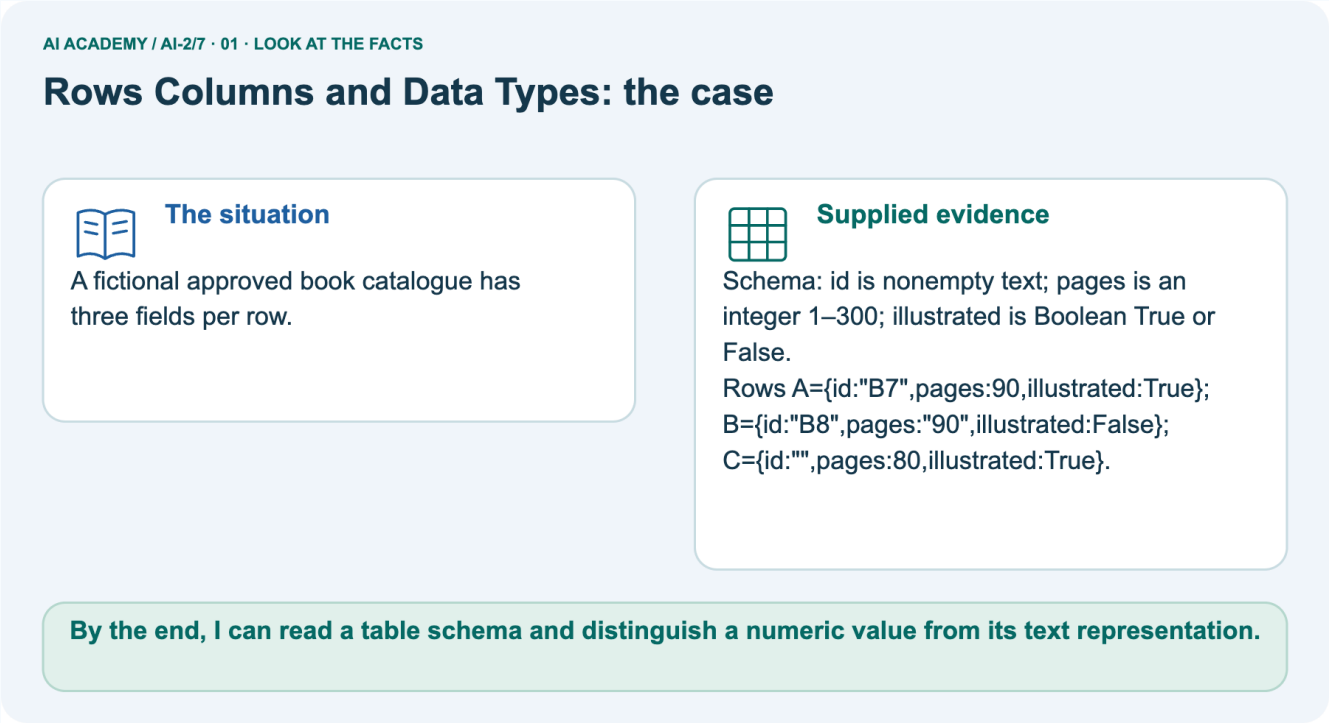
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Rows Columns and Data Types: the case



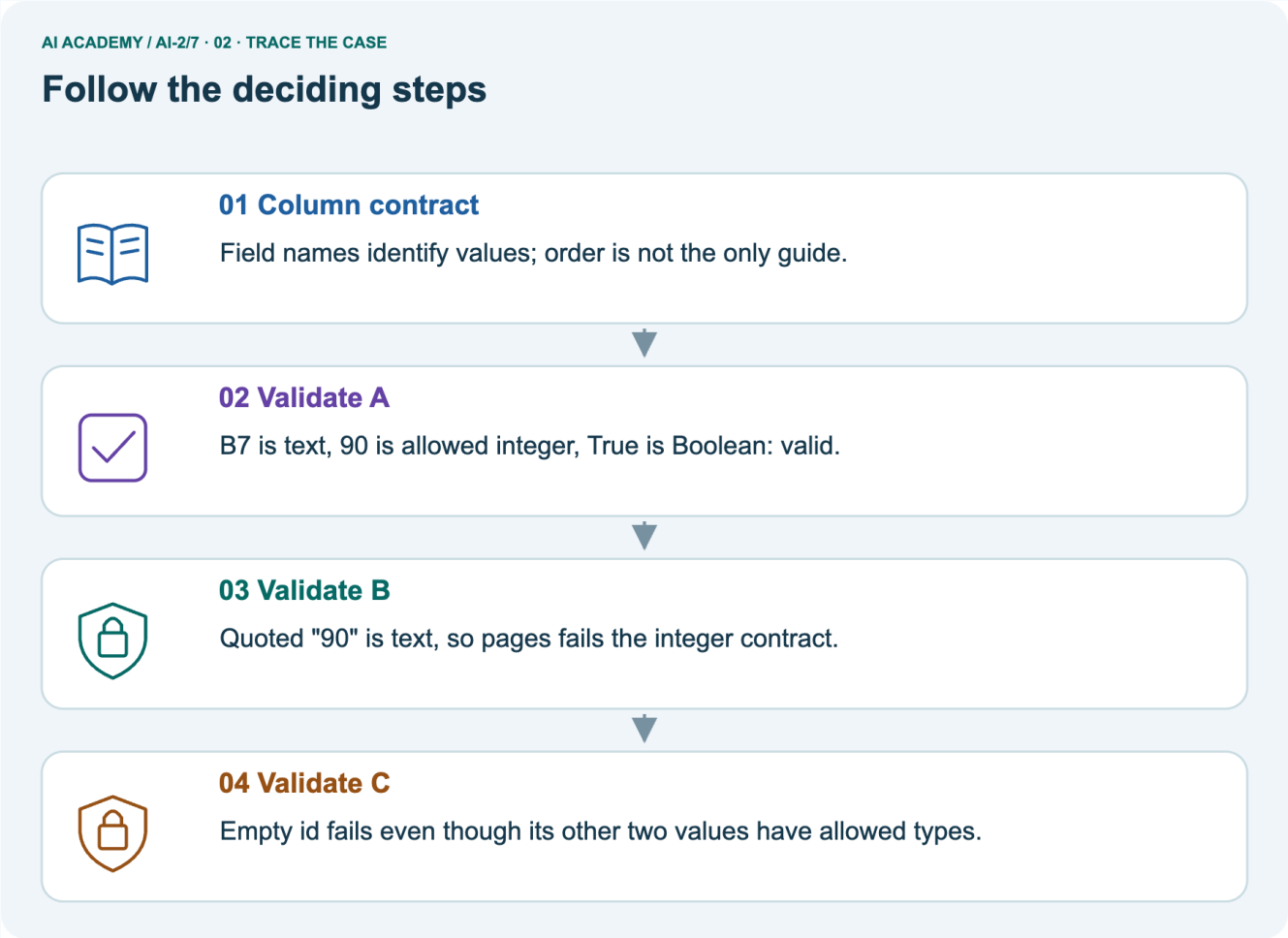
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional approved book catalogue has three fields per row. Supplied facts: **Schema**: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False}; C={id:"",pages:80,illustrated:True}.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

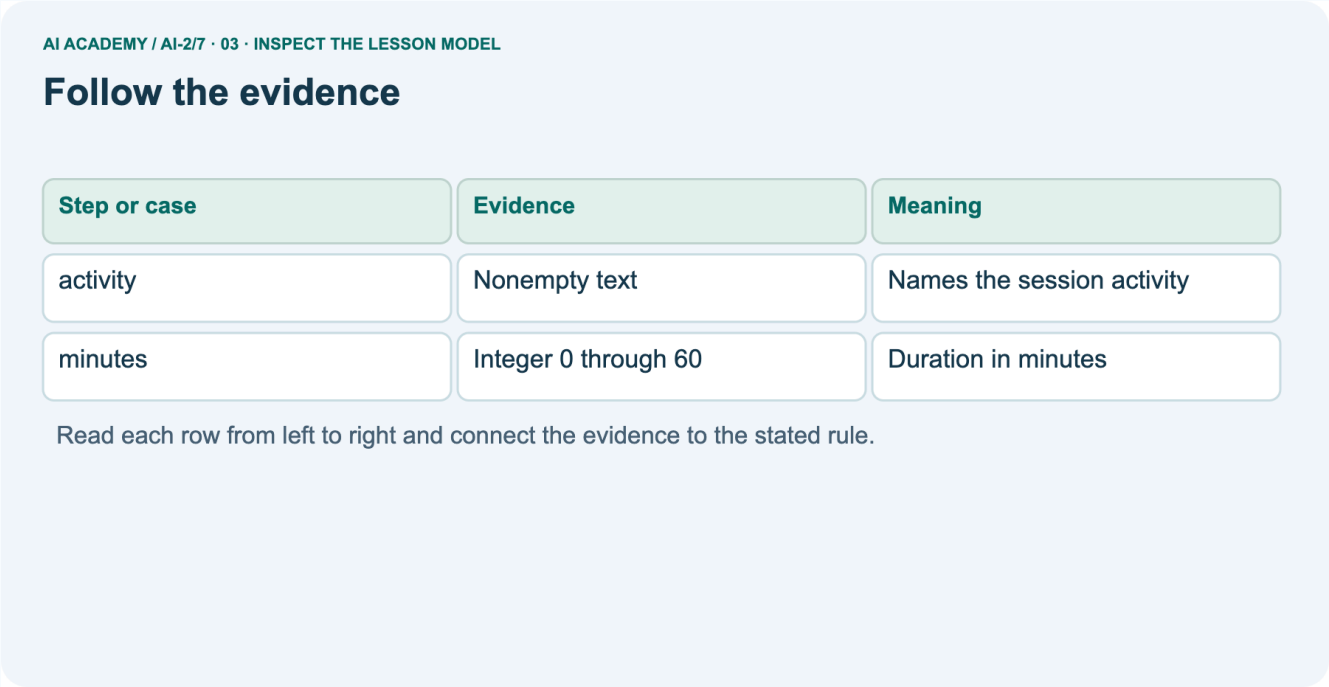


Follow the labelled picture with the matching explanation. Open the vector diagram

Column contract: Field names identify values; order is not the only guide. → Validate A: B7 is text, 90 is allowed integer, True is Boolean: valid. → Validate B: Quoted "90" is text, so pages fails the integer contract. → Validate C: Empty id fails even though its other two values have allowed **type**s.

Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

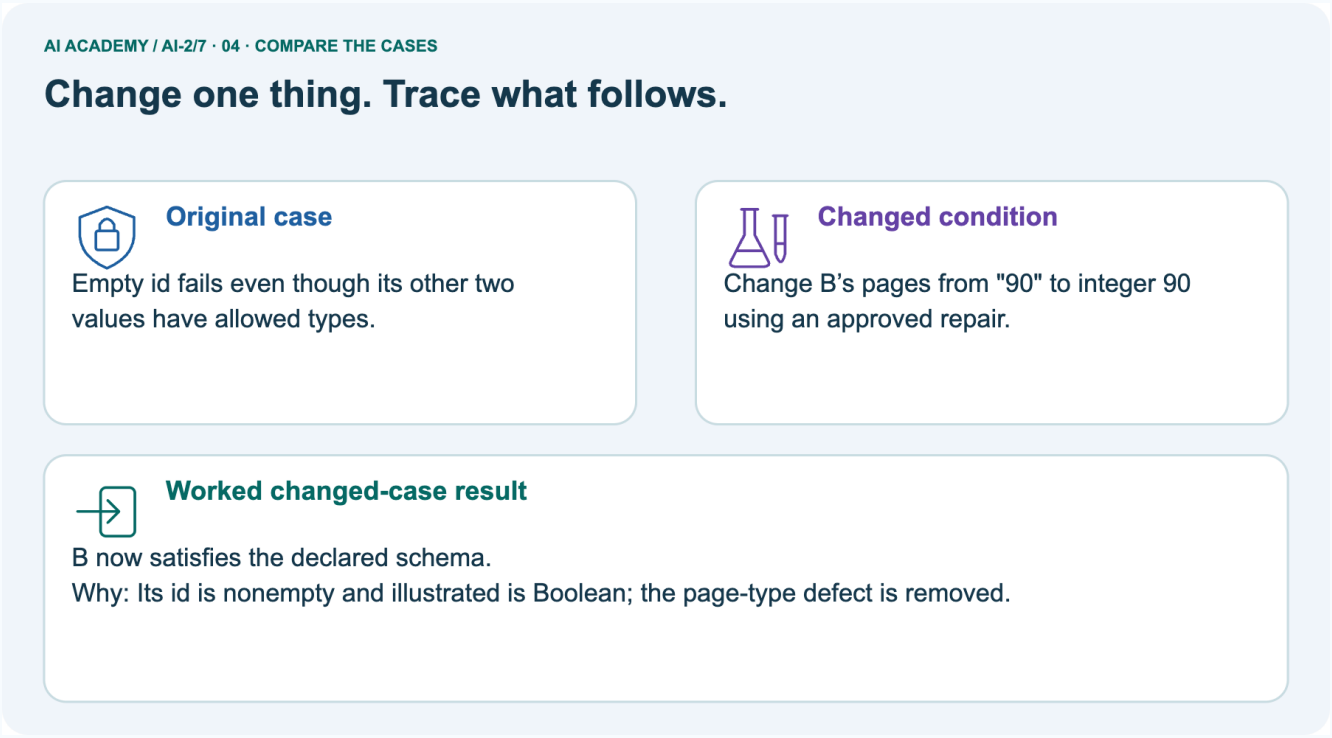


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. activity | Nonempty text | Names the session activity. minutes | Integer 0 through 60 | Duration in minutes

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



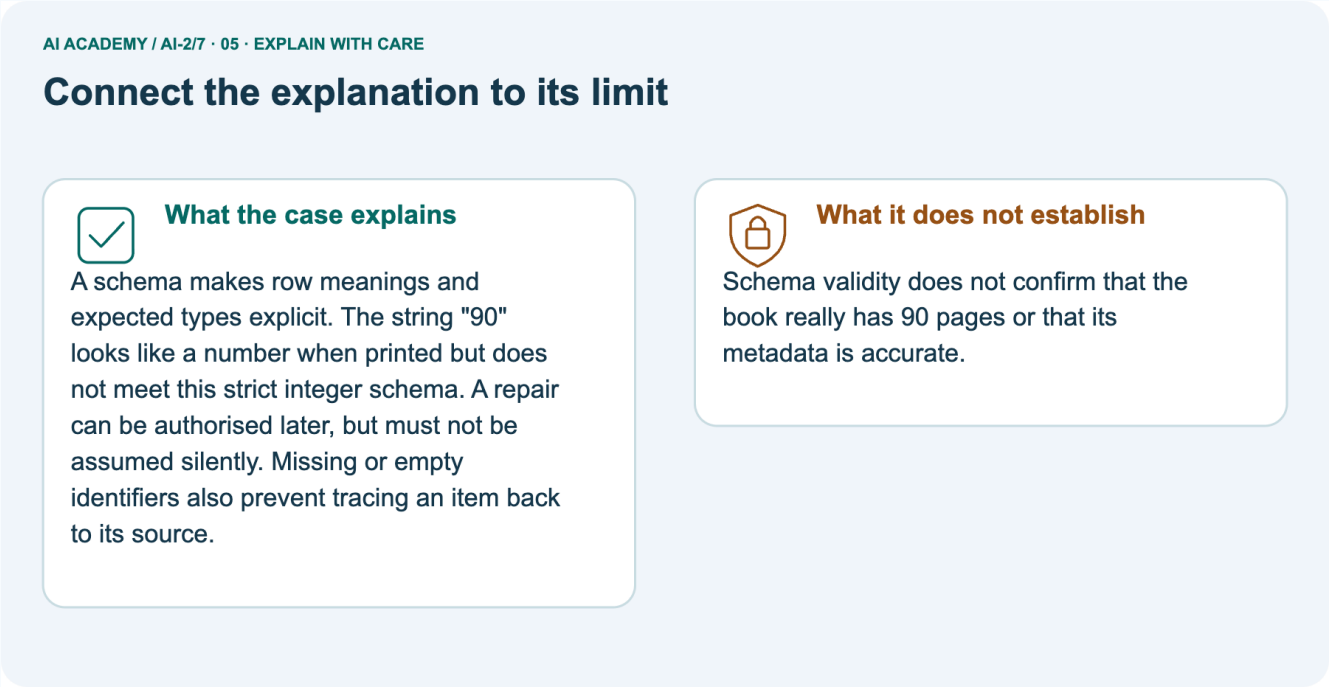
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Empty id fails even though its other two values have allowed **type**s. Changed condition: Change B's pages from "90" to integer 90 using an approved repair. Expected result: B now satisfies the declared **schema**.

Reason: Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



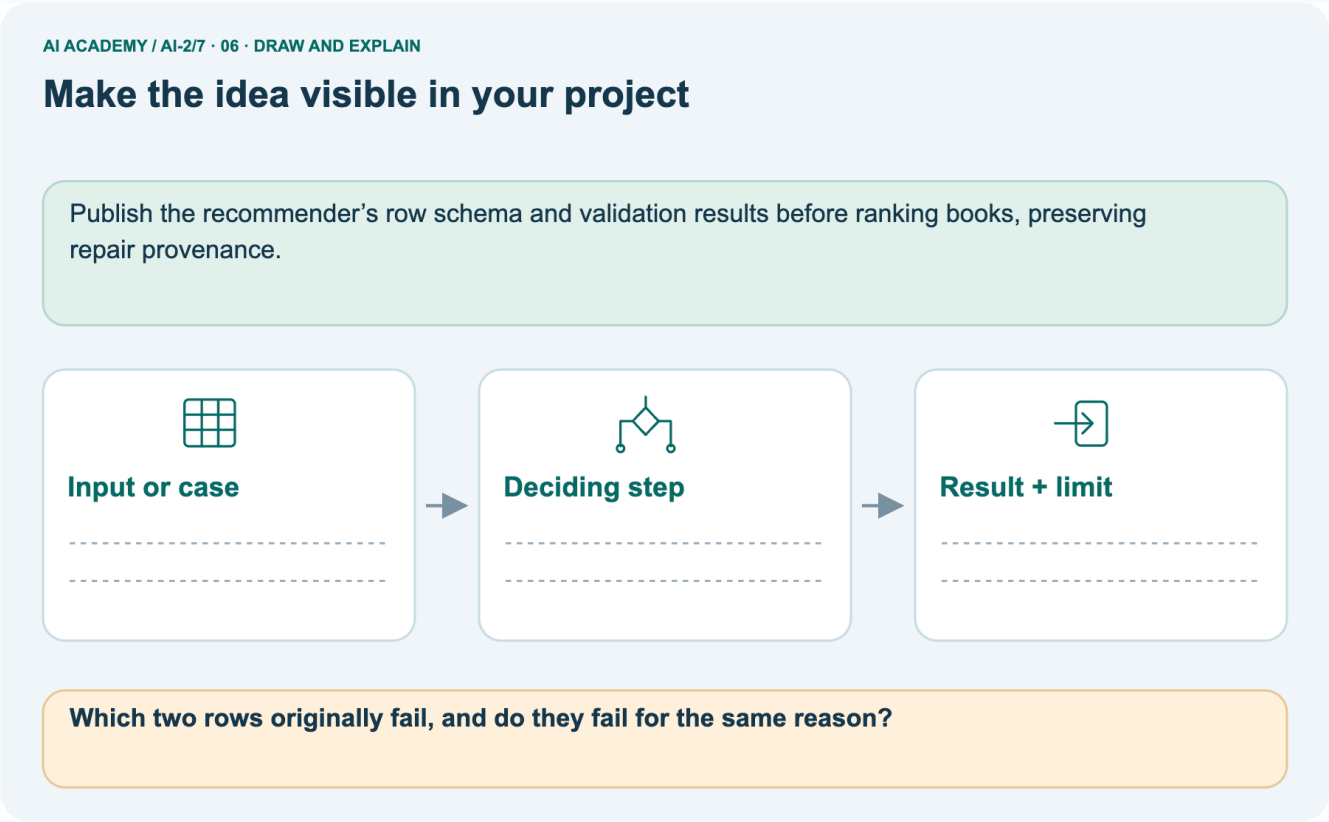
Follow the labelled picture with the matching explanation. Open the vector diagram

A **schema** makes row meanings and expected **type**s explicit. The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source. Limit: **Schema** validity does not confirm that the book really has 90 pages or that its metadata is accurate.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender's row **schema** and validation results before ranking books, preserving repair provenance. Draw your **input**, deciding step, result and limit.

Supported question: Which two rows originally fail, and do they fail for the same reason?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Rows Columns and Data Types: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.